Chapter 5
Scale Detection Using Semivariograms and Autocorrelograms

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OBJECTIVES

The evolution and ecology of all organisms are contingent on the complex variation seen in nature. Landscape ecology differs from most other branches of ecology in that it explicitly involves spatial variation. Therefore, one of the goals of landscape ecology is to describe spatial variation. The purpose of this exercise is to:

1. Introduce two tools for describing this variation: semivariance and autocorrelation; and
2. Give students experience creating and interpreting semivariograms and autocorrelograms.

In this lab, you will collect field data from quadrats arranged along a transect (or alternatively, you will use supplied data). You will then calculate and graph semivariograms and autocorrelograms using a spreadsheet, and you will use these graphs to determine how spatial patterns vary as a function of scale in your system. For this lab, you will need access to a spreadsheet program (such as Excel) and the file **vario.xlsx** provided on the book's website. If you choose the fieldwork option, you will also need two 100-m measuring tapes and one 1×1-m sampling quadrat. We've also provided code (see book's website) if you'd like to try the lab using R software.

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INTRODUCTION

Nature is intrinsically variable, and the evolution and ecology of all organisms are contingent on such variation. Landscape ecology is concerned not only with the magnitude of this variation, but also with its geometry. Most patterns in nature are far more complex than the simple polygons and curves of Euclidean geometry. For example, forest edges are rarely straight lines, animal home ranges are not rectangles, and trees are not cones. Therefore, we need special methods to describe the shape of nature.

The discipline of spatial statistics has diversified and matured (see Cressie 1991; Bailey and Gatrell 1995), and it is not possible here to give a full summary of the wealth of methods available. Instead, the purpose of this exercise is to describe two different methods for characterizing variation in a variable as a function of position in the landscape. This variable could be a soil nutrient, a measure of vegetation height, an index of species composition, or anything else of interest. In spatial statistics, we term variables with known locations **regionalized variables**, and we label them z (so as not to confuse them with x and y, typically reserved for the spatial coordinates, or for independent and dependent variables, respectively).

The two methods covered in this exercise are variography, which is part of the discipline of geostatistics (see Isaaks and Srivastava 1989), and autocorrelation, which is derived from the familiar correlation coefficient (Sokal and Rohlf 1981). Recall that the **correlation coefficient**, r, is a number that varies between −1 and +1 and reveals the nature of the relationship between two variables. It is close to −1 for two variables that are strongly negatively related, close to 0 for unrelated variables, and close to +1 for positively related variables. In contrast, variography is derived from the *variance*, which must be a positive number but can otherwise take any value.

One of the most important properties of almost all regionalized variables is **spatial dependence**. Spatial dependence (as assessed by **spatial autocorrelation**, or the tendency of a random variable to be correlated with itself at finite distances) means that a variable measured at one location *depends*, in one way or another, on the same variable measured at a different location. Spatial dependence arises for a number of different reasons, but let us consider two examples.

If you examine mean annual temperature as a function of position on the globe, you will note that (with many important and interesting exceptions) there is a gradient from warm temperatures at the equator to cool temperatures at the poles. If you have two sites that are almost at the same latitude, they will have similar temperatures. On the other hand, two sites that are on different latitudes will have different temperatures, and the amount of the difference in temperature will be positively (and gradually) related to the difference in latitude. **Spatial dependence** occurs when information available at one location allows you to infer information about the other location.

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Another example is in a savanna landscape where widely spaced trees provide islands of shade in an otherwise sunny landscape. Two sites that are centimeters apart are likely to have a similar amount of sunshine. However, two sites that are several meters apart may, or may not, have similar amounts of sunshine—a lot depends on the size and spacing of trees. If the sites are hundreds of meters apart, you may not be able to predict the sunlight regime very well. So in this case, we have spatial dependence at fine scales, but not necessarily at coarse scales. Also, unlike the example of global temperature, our regionalized variable consists of fairly discrete *patches* of sun and shade.

The first column of graphs in Figure 5.1 displays a variety of made-up regionalized variables with identical means and variances. These hypothetical variables have been constructed to illustrate the diversity of patterns that could potentially be found in nature. Note that regionalized variables can consist of a variety of features such as patches (i.e., homogeneous regions), noise (random, independent variation), random walks (a random walk is when a value at a given location equals the value at an adjacent location, plus or minus a small random number), or some combination of these. Also, note that the different variables behave differently as a function of scale. For example, patches can be large, small, or intermediate. Stretches of linear behavior can also be large, small, or intermediate. Also, noise can operate at any scale. If the graphs in Figure 5.1 were based on real data, we would seek biological explanations for the different scales. Such explanations might involve the size and shape of underlying geomorphology, the average size of plant clones, the average size of a natural disturbance, the home range size of the dominant mammal species, or the average farm size.

Except for variable A in Figure 5.1, there is some spatial dependence. That is, nearby locations are, *on average*, more similar than distant locations. Since similarity typically decreases as a function of distance of separation, we also call this phenomenon **distance decay**. Distance decay has important consequences for living things. For example, if soil conditions are very similar at nearby locations (as for variables C and F in Figure 5.1), then natural selection *might* favor plants with short dispersal distances. If, on the other hand, soil conditions were spatially unpredictable (as for variable A), a long dispersal distance *might* be advantageous. Similarly, the foraging behavior of animals, the growth of plant roots, the spread of fire, the flow of water, and the behavior of many other ecological phenomena all depend on the nature of distance decay in environmental factors.

In statistics, spatial dependence has both desirable and undesirable attributes (Legendre 1993). It means that one can predict variables (to some degree) based on geographic location, which can aid in mapping the environment. However, spatial dependence also violates the standard statistical assumption of independent observations (even if samples are randomly located). Thus, unless specifically corrected for, many statistical methods are invalid if your data exhibit distance decay. Fortunately, there are tools to evaluate the *degree* and the *scales* of spatial dependence. The two tools we introduce in this laboratory exercise are the semivariogram and the autocorrelogram.

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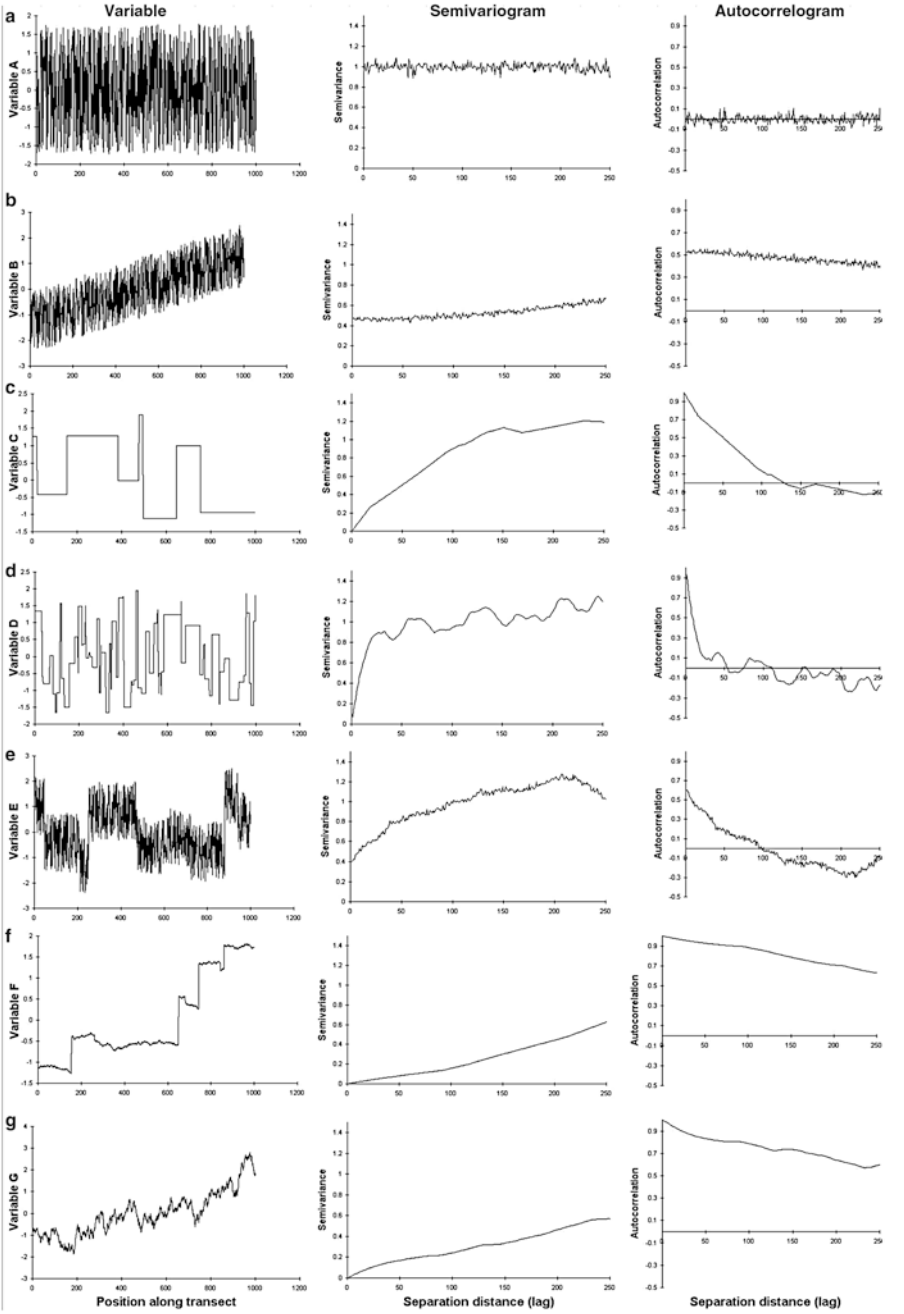


Figure 5.1 Seven artificial regionalized variables (column 1) as a function of position along a transect, along with their corresponding semivariograms (column 2) and autocorrelograms (column 3). All variables have identical means and variances. The variables can be described as follows: (**a**) pure noise; (**b**) fine-scale noise superimposed on a linear trend; (**c**) large patches; (**d**) small patches; (**e**) noise superimposed on large patches; (**f**) patches with "drift" in their mean values, plus fine-scale noise; (**g**) random walk

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Variography

Variography is the discipline of using semivariograms (and related graphs such as covariograms) to uncover the degree to which the variance in a regionalized variable depends on distance (Rossi et al. 1992). The geographic distance between two samples is termed the **spatial lag**. Recall that the variance is the square of the standard deviation and is a measure of the spread or variation of data. The word **semivariance** is derived from "half of the variance," and indeed it is a measure of the variance of the regionalized variable, z. But what is special about the semivariance is that it changes as a function of distance. The semivariance is computed as follows:

$$\gamma(h) = \left\{\Sigma\left[z(i) - z(i+h)\right]^2\right\} / 2N(h)$$

where $\gamma(h)$ is the semivariance of a **lag** of distance h, $z(i)$ is the value of a regionalized variable z at location i, $z(i+h)$ is the value of z at a location separated from i by lag h, and $N(h)$ is the number of pairs of points separated by lag h. The summation is over all pairs of points separated by distance h. In plain English, the semivariance is half of the average squared difference of all pairs of points separated by a given distance. A semivariogram is a plot of semivariance versus the lag distance. As with the variance, the semivariance cannot be less than zero, but it is not bounded on the top.

An idealized, hypothetical semivariogram is given in Figure 5.2. Since the semivariance is directly related to variance, a high value indicates high variation, and a low value indicates low variation. Almost always, variance increases as a function of lag

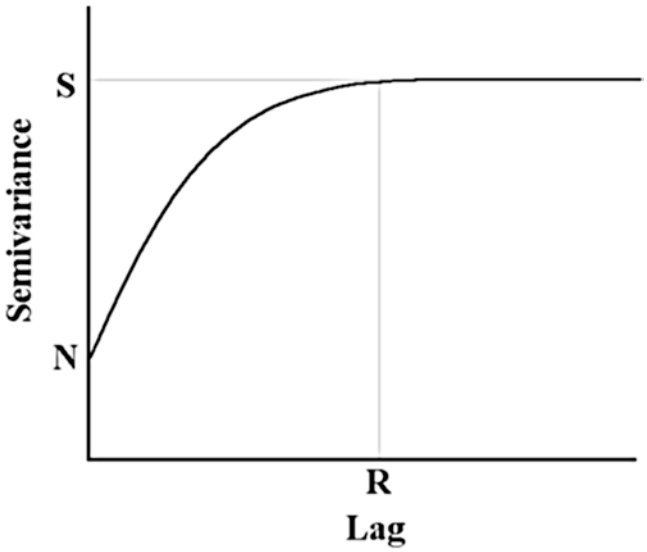


Figure 5.2 An idealized semivariogram. N = nugget, R = range, S = sill

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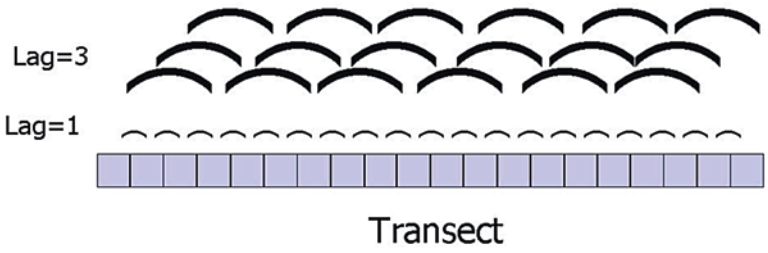


Figure 5.3 A 20-quadrat-long transect illustrating all of the pairs of points separated by two selected lags: 1 and 3

distance. In other words, the larger the area you study, the more variable your conditions are. It is important to reiterate that the lag distance is not the same as the distance from the origin or starting point. Rather, it is calculated for all pairs of points (Figure 5.3).

At distances less than R (the **range**), we have spatial dependence (Figure 5.2). That is, closer samples are more similar than distant samples. At distances of at least h, we have spatial independence; therefore, samples separated by longer distances would be valid for conventional statistics. Any area with linear dimensions of at least R would have as much variance as the landscape as a whole. A horizontal asymptote at distances greater than R is known as the **sill**. The sill indicates the amount of "background" variation.

For a very smooth, regionalized variable, two samples that are infinitesimally close to each other may have almost identical values. Elevation of the ground surface almost always behaves this way. However, most variables are not so smooth, if for no other reason than measurement error. Such unresolved variation at very fine scales is termed the **nugget effect**, and is indicated by N (Figure 5.2). The term derives from the original use of variography in gold mining: at fine spatial scales, you either find the gold nugget or miss it. Soil variables such as pH or nutrient concentrations typically have very high nugget effects.

The semivariogram specifically address how variance increases as a function of scale. Although it can only describe *patterns*, we often hope to infer the *processes* that generate such patterns. If we find a distinct range, or even a pronounced inflection in the semivariogram, we suspect that there are different processes operating at different scales. For example, if we discover that the range equals approximately 10 m, we need to seek an underlying process that operates on a scale of 10 m. In a forest, this scale could represent the average canopy gap size or the average size of a canopy tree crown. In the arctic tundra, the range could represent the average size of a permafrost polygon. Of course, you are never guaranteed to actually find a range. It is possible (and indeed, likely) that variation in nature increases continuously as a function of scale.

The second column of Figure 5.1 shows the semivariograms for the hypothetical regionalized variables. Since very few pairs of points represent very far distances, it is usually not advisable to plot $\gamma(h)$ for large h. A general rule of thumb (adopted here) is to plot only up to half of the maximum distance between samples. The

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bumps and wiggles at such far distances in the semivariograms are due to chance variation in the data, and not to the underlying process generating the patterns. Note that only three of the variables (C, D, and E) have semivariograms remotely resembling those in Figure 5.2. The range of variable C (≈150 units) is much larger than the range of variable D (≈25 units), which reflects the differences in patch size. Variable E seems to have an inflection at 50 m, which marks the difference between noise within patches, and the differences between patches.

The semivariograms of variables B, F, and G are continuously increasing functions. Therefore, there is spatial dependence at broad spatial scales. Three variables (A, B, and E) have much fine-scale noise, and hence a substantial nugget effect. Note that variable F has very little fine-scale noise, and hence has a negligible nugget. Variable A represents pure noise and hence pure spatial independence. For such variables, the nugget equals the sill.

Autocorrelation

Autocorrelograms are plots of the correlation coefficient, *r*, as a function of lag:

$$r(h) = \mathrm{corr}\left[z(i), z(i+h)\right]$$

It is called "auto"-correlation because the variable is correlated with itself. Autocorrelation can take values from −1 to +1 although for most applications positive values are most common. In situations with distance decay, autocorrelograms are declining functions and often look like upside-down semivariograms (third column of Figure 5.1). If there is little fine-scale noise, the *y*-intercept will be close to 1.0. In situations in which the semivariogram displays a nugget effect, the *y*-intercept of the autocorrelogram will be less than 1. An autocorrelation of 0 means there is no spatial predictability; this is related to the concept of the sill.

This describes only the simplest kind of autocorrelogram. More complex (and usually more appropriate) ways to calculate autocorrelograms, as well as testing their statistical significance, are described by Legendre and Fortin (1989), Bailey and Gatrell (1995), and Legendre and Legendre (1998). Autocorrelograms are also used in the analysis of change through time (also known as "time series").

Comparing Autocorrelation to Semivariance

The interpretation of autocorrelograms is very similar to that of semivariograms, so the choice between them is largely a matter of taste. Since the correlation coefficient is a dimensionless number (i.e., it is standardized), autocorrelograms are useful in comparing variables with different units (e.g., plant density and soil calcium). Semivariance has a dimension of units squared (so if the regionalized variable is in

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parts per million or ppm, semivariance is in ppm^2). Thus, it is useful in comparing different commensurate variables or (more commonly) the same variable in different locations. However, semivariance can be standardized for comparing variables measured in different units (see Rossi et al. 1992).

Since it is derived from the correlation coefficient, autocorrelation is closely related to classical statistical theory. Variography, on the other hand, is a branch of geostatistics. This discipline was largely developed for the mining industry to help predict the locations of mineral deposits. Variography is a precursor to geostatistical interpolation (for mapping) or "kriging" (see Isaaks and Srivastava 1989) and to fractal geometry (Burrough 1983; Palmer 1988).

EXERCISES

Use Option 2: Provided Datasets
We won't be able to collect our own data, but it may be useful to read this for info on how these kinds of data are collected.

EXERCISE 1: Data Collection

Option 1: Field Exercise Using Vegetation Height

1. Choose a field site in which the maximum height of the vegetation is about 2.5 m or less, and in which it is possible to fit a 200-m transect. If the site is large enough, randomly choose a starting location and compass direction. If it is too small for this, choose an appropriate direction but randomize the starting point, so you are not biased by particular plants.
2. Extend two (or more) 100-m tapes end to end along the chosen compass direction. Ideally, you would do this with a surveyor's compass or level. It is crucial that the transect be as straight as possible and not influenced by the vegetation!
3. Beginning at 0 m, establish a 1×1-m quadrat (this can consist of three meter sticks plus the meter tape as the fourth boundary).
4. Within this plot, measure and record the height of the tallest plant.
5. Now repeat the process with an adjacent plot at 1 m, then at 2 m, and continue to the end of the transect.

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